

56.6%
LATE DELIVERY RATE

-46.6%
SLA DEFICIT (VS 90%)

77.4%
DELAY IS PRE-PICKUP

4.25%
CANCELLATIONS (68 ORDERS)

1. CLAIM EVALUATION: "TRAFFIC IS THE PROBLEM" REFUTED

- Refuted by Data:** 50.1% of orders are late in *low traffic*, and 53.6% fail in *clear weather*. Low/Med traffic accounts for 63.4% of total volume.
- Distance is Non-Significant:** Orders under 5 km fail at 55.6% (Chi-square p-value = 0.419). Traffic adds ~3 minutes to severe trips but does not cause baseline failures.

Traffic Level vs. Late Delivery Rate

Traffic Level	Late Rate (%)
Low	50.1%
Med	57.6%
High	66.0%
Severe	65.6%

2. TIMELINE DECOMPOSITION: WHERE TIME IS LOST 77% PRE-PICKUP

- Pre-Pickup Blowout:** 77.4% of excess delay (+14.2 min) occurs before pickup ('created_at' to 'pickup_at'). Road transit expands by only +4.1 min (22.6%).
- Customer Tickets:** 49.3% cite kitchen/courier delays; 18.4% cite dynamic ETA creep.

Delay Breakdown: 77% is Pre-Pickup

Category	Excess Delay (Minutes)	Percentage
In-Transit (Road Travel)	+4.1m	23%
Pre-Pickup (Kitchen & Dispatch)	+14.2m	77%

3. CONTRACTUAL SLA PERFORMANCE 43.4% ON-TIME

- Census Population:** Strictly N = 1,532 valid delivered orders (3 DB duplicates removed; 37 missing timestamps recovered).
- Severe SLA Breach:** Actual on-time rate is 43.4% against the 90.0% contractual commitment (-46.6% gap).
- Delay Magnitude:** Median late order is +7.9 min late (mean +13.6 min; tail reaches +50.3 min).

On-Time vs. Contract SLA Target

Metric	Value (%)
Actual	43.4%
SLA Target	90.0%

4. TELEMETRY BLINDSPOT & GOVERNANCE ATTRIBUTION GAP

Delivery Process & Missing Measurement Point:

Placed ➤ Assigned ➤ **? Arrived ?** ➤ Picked Up ➤ Delivered

- Zero Arrival Telemetry:** No record exists of when drivers arrive at restaurants ('driver_arrived_at_restaurant' is missing).
- Coarse GPS Pings:** Pings occur every 13.4 minutes on average (wider than typical 5–10 min restaurant dwell).
- Attribution Gap:** We cannot determine if food was waiting for couriers, or if couriers were waiting for food.

Data Source	Reliability	Audit Finding
orders (SQLite)	Medium	3 duplicate rows; 37 missing drop-offs.
driver_events.json	High	Authoritative ground truth for drop-offs.
Dispatch API	High	1,600/1,600 verified; tracks ETA creep.
restaurant_status	Untrusted	31% coverage; manual staff clicks.

5. ACTION PLAN NEXT STEPS

Priority	Action Item	Owner
P0	Geofenced Arrival: Log automatic 'driver_arrived' ping within 100m.	Mobile Eng
P0	Recalibrate ETA: Add +12 min baseline buffer to pre-pickup ETAs.	Product Ops
P1	POS / KDS Integration: Capture automated cooking start and ready pings.	Merchant Eng

6. Decision: Do Not Build AI Delay Predictor

An ML model trained on current data will confuse kitchen delays with courier delays, creating biased predictions and unfair operational penalties. Fix physical arrival telemetry before building AI models.